Lesson notes · for you

Inside an agent

AI agents · Lower secondary · Lesson 1

Time
20 min

Materials
Paper and a pen; A timetable, a bus timetable or a weather forecast (from the radio, TV or a newspaper); A friend or someone at home to play the tool

Learning goals

- Describe the agent loop.
- Read a tool call: its name and its details.
- Put a short Python loop in order.

Activity: Be the agent

1. Warm-up (3 min): List five apps you or your family use. Next to each, write “talk” if it only writes text back, or “act” if it does something, such as setting an alarm or finding a route.
2. Write (7 min): Choose three requests, such as “What’s the weather in our town tomorrow?”. For each, write the tool call: the tool’s name, then the details in curly brackets.
3. Play (7 min): With a partner, one of you is the model and the other the tool. The model hands over each call without saying the answer; the tool writes the result from a timetable or forecast; the model answers from that result only.
4. Check (3 min): Read your answers again. Did each one use the result, or did the model guess? Swap roles and play one more round.

You only need paper, a pen, and a timetable or a weather forecast. No phone is needed.

Discussion

- Which apps on your phone are agents, and which only talk?
- Why is it useful that the app, not the model, runs the tools?
- What would you want to check before an agent acts for you?

Check yourself

- Write one tool call for a request of your own: the tool’s name and its details.

Try it this week
Write down three things you asked a phone or an app to do this week. For each one, decide: only talk, or a tool? Then write the tool call it might have made, with a name and details.

Worksheet

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Name _____ Class _____ Date _____

1 **On the trip from Kafi to Dala, how many tools does Darsi Tiny call before it answers?**

○ Five ○ One ○ Two

2 **Put the agent loop in order.**

☐ The model writes a plan

☐ You make a request

☐ The model answers from the result

☐ The model writes a tool call

☐ The app runs the tool and adds the result

3 **What does this program print?**

```
call = 'weather{"place":"Kafi"}'
name = call.split("{")[0]
print(name)
```

○ place ○ weather ○ Kafi

4 **This app's loop uses a scripted list in place of a real model. Put the lines in order (with their indents) so it runs a tool for each call and stops at the answer.**

Number the lines in order (1, 2, 3...), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `break`

☐ `replies = ["call", "call", "answer"]`

☐ `for reply in replies:`

☐ `if reply == "answer":`

☐ `print("App runs a tool")`

☐ `print("Model runs a tool")`

☐ `print("Done: answer")`

5 **Match each part of the trip from Kafi to Dala to what it was.**

Plan	• • • •	route, from Kafi to Dala
First tool call	• • • •	"First the route, then the weather."
First result	• • • •	"From Kafi to Dala: 70 km... Tomorrow in Dala: partly cloudy..."
Answer	• • • •	70 km

✎ **Think of an app on your phone that does things for you. What tools might it use, and what would you want to see before it acts?**

Check your answers

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1 **Two**

The route tool first (70 km), then the weather tool for Dala, and then one answer with both.

2 **Top to bottom: 2, 1, 5, 3, 4**

The model plans before it calls a tool. Only the app can run the tool and add the result, and only then can the model answer from it.

3 **weather**

split("{") cuts the call at the curly bracket, and [0] takes the first piece: the tool's name, weather.

4 **Top to bottom: 5, 1, 2, 3, 6, X, 4**

```
replies = ["call", "call", "answer"]
for reply in replies:
    if reply == "answer":
        print("Done: answer")
        break
    print("App runs a tool")
```

First the list, then the loop. Inside it, the if checks for the answer and break stops the loop; otherwise the app runs a tool.

5 **Plan — "First the route, then the weather."**
First tool call — route, from Kafi to Dala
First result — 70 km
Answer — "From Kafi to Dala: 70 km… Tomorrow in Dala: partly cloudy…"

The plan names the steps, each call names a tool and its details, each result comes back from the app, and the answer uses both results.